25901742 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCMOS100 - FOUNDATIONS OF MOBILE DEVELOPMENT SYSTEMS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. What is the function of && operator in Dart? [U] / [CO1]

   a). To perform a logical AND operation

   b). To perform a logical OR operation

   c). To perform a logical NOT operation

   d). To perform an arithmetic operation

2. Which collection uses key-value pairs? [U] / [CO1]

   a). List b). Set

   c). Map d). Queue

3. What is the correct syntax to declare a map in Dart? [U] / [CO1]

a). var map = {}; b). var map = [];

c). var map = {'key': 'value'}; d). var map = ('key', 'value');

4. Which method is used to save a string value in Shared Preferences? [K] / [CO2]

a). setstring() b). saveString()

c). storeString() d). putString()

5. If running flutter doctor show issues what is likely first step to resolve them? [U] / [CO2]

a). Reinstall flutter b). Run flutter clean

c). Restart the IDE d). Follow the recommendations provided by flutter doctor

6. What is the shortcut key to open the command palette in V SCode? [K] / [CO2]

a). Ctrl + Shift + P b). Ctrl + Shift + I c). Ctrl + Shift +O d). Ctrl + Shift + K

7. What is the main advantage of using Firestore over Firebase Realtime Database? [U] / [CO3]

a). Firestore is better for large binary files b). Firestore supports offline data persistence c). Firestore allows for more flexible querying and scaling d). Firestore is cheaper to use

8. What programming language is used to develop Flutter applications? [U] / [CO3]

a). Java b). Kotlin c). Swift d). Dart

9. What is Firebase Authentication mainly used for? [U] / [CO3]

a).Sending emails b).Storing images c).Managing user sign-in and sign-up d).Building UI

10. What is the purpose of the PATCH method in RESTful APIs? [K] / [CO3]

a). To completely replace a resource b). To apply partial updates to a resource c). To delete a resource d). To retrieve data from a server

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11.	How do try, catch, and finally work in Dart for handling exceptions? Provide a real-world Flutter example.	[U] / [CO1]
12.	What is abstraction in Dart, and how does it simplify complex Flutter applications?	[U] / [CO1]
13.	Which Flutter widget library is specifically designed to replicate iOS-style UI components?	[K] / [CO2]
14.	Distinguish between the release mode and build mode.	[U] / [CO2]
15.	Explain the concept of Firestore in flutter.	[U] / [CO3]
16.	Describe the role of SQLite in Flutter applications.	[U] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17.	Discuss the importance of objects in managing application data in Flutter. How do they interact with UI components?	[U] / [CO1]
18.	Compare and contrast different looping mechanisms in Dart (for, while, do-while)	[U] / [CO1]
19.	Discuss the differences between Material and Cupertino widgets in Flutter and their appropriate use cases.	[U] / [CO2]
20.	What is the difference between idempotent and non-idempotent HTTP methods in RESTful APIs?	[K] / [CO3]

[10x2 = 20]